

Capita Selecta – Project Plan

Comparing Bluetooth, Li-Fi and Wi-fi for Sending Sensor Data

Saxion University of Applied Sciences

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Abbreviations

Term	Meaning
Li-Fi	A method that sends data through light, using a LED as the transmitter and a photodiode as the receiver
Wi-Fi	A common wireless method that sends data through radio waves, often used for internet access
BLE	Bluetooth Low Energy, a short-range wireless method that sends data through radio waves and uses little power
Raspberry Pi Pico 2 W	A small, low-cost microcontroller board with built-in Wi-Fi and Bluetooth
MQTT	A lightweight messaging method used to send data from one device to another, often used for sensor data
KMP	Kotlin Multiplatform is a way to write shared code once and run it on several platforms, such as Android, iOS, web, and desktop.
OOK	On/off keying is a way to send data by turning a signal, such as a light, on and off.
BER	Bit Error Rate, a measure of how many bits are received wrong compared to how many were sent
INA219	A small sensor chip that measures voltage and current, used here to measure power use
Internal temperature sensor	A sensor built into the Raspberry Pico 2 W chip itself, which measures the chip's own internal heat. It rises when the chip works harder, for example, while sending data over Wi-Fi, BLE, or Li-Fi, so it works as a sign of how much load each method places on the chip
Native	In this document, a native chart or a native notification means a tool built specifically for one platform, such as Android or iOS, rather than a shared tool used across all platforms. Each platform draws its own charts and shows its own notifications with its own separate library.
EC	European Credit, a unit that shows how many study hours a course requires
GPIO	General Purpose Input/Output
PWM	Pulse Width Modulation

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1. Introduction

Figure 1 depicts a Raspberry Pi Pico 2 W reading its internal temperature sensor and transmitting the same value over Li-Fi, BLE, and Wi-Fi at the same time to a Raspberry Pi 4, which acknowledges data back over the same channel, logs and benchmarks the results, and publishes them via MQTT to a Kotlin Multiplatform app.

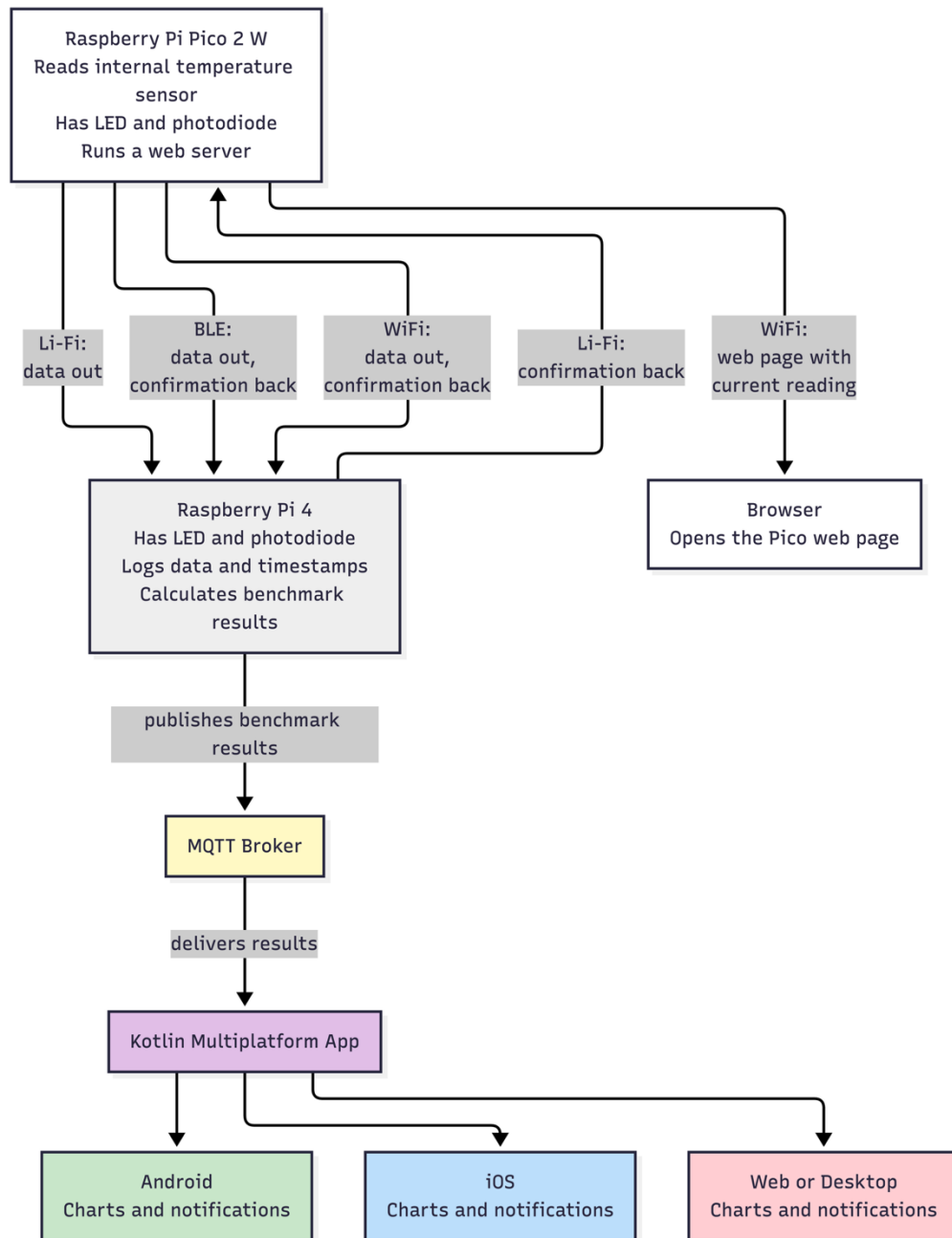


Figure 1- Overview of the sensor data path from the Raspberry Pi Pico 2 W to the Kotlin Multiplatform app

Two radio technologies dominate wireless sensor communication: Bluetooth and Wi-Fi. Bluetooth, including its variant Bluetooth Low Energy (BLE), has existed for 25 years and appears in most computing devices (Zeadally et al., 2019). Wi-Fi has moved through generations from 802.11b to the coming Wi-Fi 8. Each generation raised throughput, reliability, and latency and added mechanisms for energy saving and coordination between access points (Geraci et al., 2026). Both technologies serve sensor data because their behavior is documented.

By competing for reliability, power consumption, and latency with each other. Research on time-slotted channel hopping in wireless sensor networks shows that improving one of these three indicators worsens the other two (Scanzio et al., 2020). This project measures several metrics together: throughput, range, reliability, latency, and power. This is the basis of the benchmark approach used in this project.

Li-Fi (optical wireless communication) appears rarely in student projects. Industry uses it. The IEEE Std 802.15.13 standard for optical wireless communication in industry reports links latency below 4 ms over several meters, without a radio license and with reduced interference (Bober et al., 2024). Hospitals, aircraft cabins, and industrial sites use links like this because radio spectrum is restricted or untrusted in these places. This project treats Li-Fi as a third option and measures it against BLE and Wi-Fi under the same conditions.

Purpose of the assignment

This project builds and tests three wireless methods: Li-Fi, BLE, and Wi-Fi. Each method sends the same sensor reading from a Raspberry Pico 2 W to a Raspberry Pi 4 at the same time. This allows a comparison between the three methods. The purpose is to answer this research question:

Under which conditions of distance, interference, and obstruction does each Li-Fi, BLE, and Wi-Fi perform best for sending sensor data, based on throughput, range, reliability, latency, and power use?

The assignment produces an answer to this question and a working system: a sensor link that works in both directions on all three media, a benchmarking pipeline, and a Kotlin Multiplatform app that shows the results live.

Methodological approach

The project follows the V model. In this method, each stage on the left side of the V pairs with a verification or validation stage on the right side. The two sides meet at implementation, at the bottom (Figure 2). This structure matches the four project phases used throughout this plan (Chapter 3). Testing verifies requirements and analysis through system testing and acceptance testing: the live demonstration and the benchmark results. Testing verifies functional and technical design through unit testing and integration testing: the codec tests, the benchmark calculation tests, and the MQTT app tests in Section 6.2. Every requirement and design decision links to a specific test.

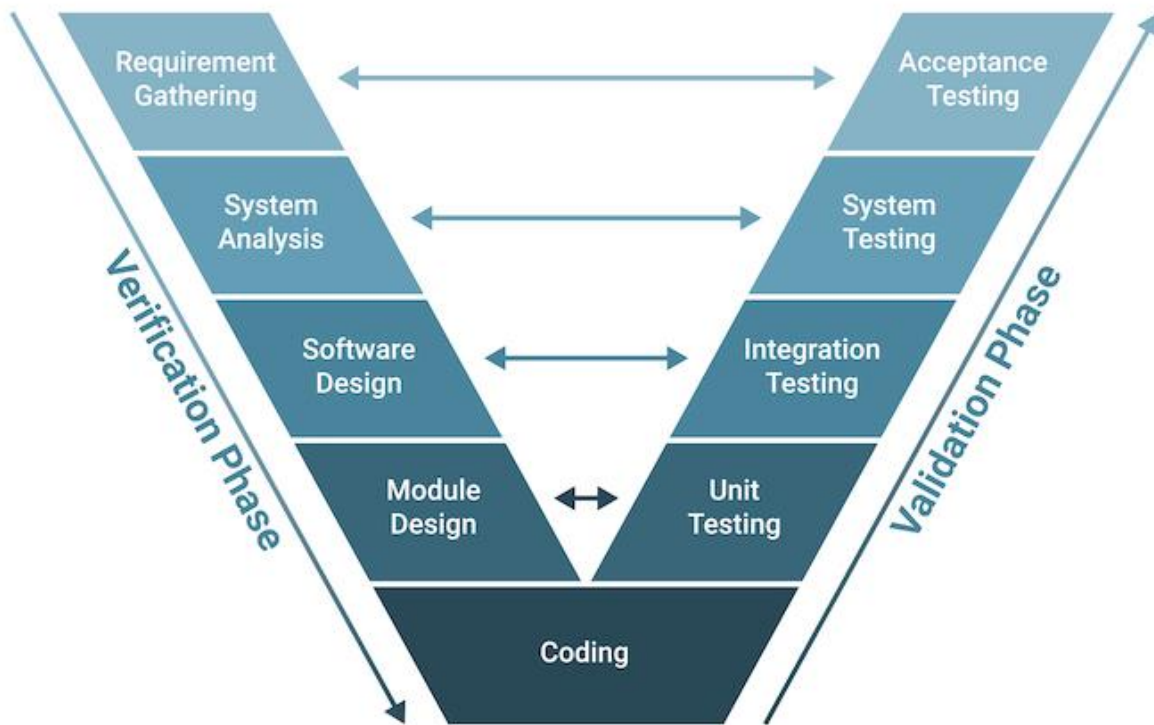


Figure 2 - The V model

The project combines electrical engineering work and computer science work:

- One sensor node, the Pico 2 W, sends the same internal temperature reading over Li-Fi, BLE, and Wi-Fi at the same time. This gives all three methods the same payload and timing for comparison.
- Each method is tested as a full-round trip. The Raspberry Pi 4 sends confirmation back over the same channel that carried the data. For Li-Fi, this return path also uses light.
- Running all three methods together makes it impossible to link a temperature or power change to one method. Power and heat are measured separately, one method at a time. Results are checked against INA219 power readings.
- Benchmarks repeat under different distance, interference, and obstruction conditions. Interference includes radio congestion for BLE and Wi-Fi and ambient light or a blocked light path for Li-Fi. Each condition runs multiple trials under one test protocol.

Chapter 3 details this phased approach: requirements and analysis, design, realization, and testing.

Report outline

Chapter 2 states the main objective and sub-objectives in SMART terms. Chapter 3 describes the project phases, the activities, and the estimated hours for each. Chapter 4 sets the

boundaries of the project with MoSCoW. Chapter 5 lists the hardware, software, and documentation of products. Chapter 6 describes how each product category is checked for quality. Chapter 7 covers project organization and communication. Chapter 8 gives the planning, detailed further in Appendix A. Chapter 9 analyses technical risks and planning risks and states how each is mitigated.

2. Project Objectives

Main objective

Design, build, and benchmark a sensor node that sends the same sensor reading through Li-Fi, Bluetooth Low Energy, and Wi-Fi from a Raspberry Pi Pico 2 W to a Raspberry Pi 4. Determine which method performs best under different conditions of distance, interference, and obstruction, measured by throughput, range, reliability, latency, and power use.

Sub-objectives

- Design and build a Li-Fi link that works in both directions, using an LED, a photodiode, a transimpedance amplifier, and a comparator stage, on both the Pico 2 W and the Pi 4.
- Implement BLE and Wi-Fi firmware on the Pico 2 W. Each method receives confirmation back from the Pi 4.
- Build a Raspberry Pi 4 pipeline that timestamps, logs, and benchmarks the data, then publish results through MQTT.
- Develop a Kotlin Multiplatform application with shared data models, an MQTT client, and benchmark logic. Add charts and notifications for each platform.
- Implement a web server on the Pico for live verification of the current reading.
- Design and run a benchmark protocol that covers distance, interference, and obstruction. Measure power and temperature for each method in isolation.
- Produce a report with source code, firmware, circuit schematics, analysis, and a live demonstration.

Execution of the project

Van Dung Phan executes the project alone, under supervision of Mr. Christiaan Slot at Saxion University of Applied Sciences. The module is Capita Selecta, 10 EC, 280 hours, over 20 weeks. Active work is planned for the first 16 weeks. The final 4 weeks are held as buffer and exam weeks. Work proceeds in four phases: requirements and analysis, design, realization, and testing. Progress is reported to the supervisor by email every week (Section 7.2). A meeting is scheduled only for a problem, a question, or a finding for worthwhile discussion. Scope is adjusted with the MoSCoW boundaries in Chapter 4 when needed.

SMART

Criterion	How this project satisfies it
Specific	Build and benchmark three transmission paths, Li-Fi, BLE, and Wi-Fi, between the Pico 2 W and the Pi 4. Measure five metrics: throughput, range, reliability, latency, and power.
Measurable	The Pi 4 pipeline logs each metric as a number: kbps, meters, bit error rate, milliseconds, and milliwatts. Results are checked against INA219 readings and the internal temperature sensor.
Acceptable	The supervisor, Mr. Christiaan Slot, reviewed and accepted the project idea and scope. The project fits the learning goals of the Capita Selecta module.
Realistic	Scope is set by a 280-hour estimate for each activity (Section 3.2). The project uses components available to buy and reuses KMP and the embedded experience the student already has.
Time-bound	The module runs for 20 weeks. Active work finishes in week 16, with weeks 17 to 20 held as buffer and exam weeks. Milestones mark each phase (Chapter 8), inside a 280-hour budget.

3. Project Activities

Phases

These four phases follow the V model from Section 1.3 (Figure 2). The two design phases, 3.1.1 and 3.1.2, each match a verification phase on the test side, 3.1. 4. The phases meet at realization and implementation, 3.1. 3.

3.1.1 Project Setup: Requirements and Analysis

Review literature on Li-Fi, BLE, and Wi-Fi comparisons, OOK modulation, and benchmark methodology (see References). Define the benchmark metrics, the confirmation protocol, and the test conditions: distance, interference, and obstruction.

3.1.2 Design Phase: Functional Design and Technical Design

Design the photodiode receiver: a transimpedance amplifier and a comparator. Design the LED driver. Design the MQTT message and topic structure. Design the architecture for the Kotlin Multiplatform app and the Pi 4 pipeline.

3.1.3 Realization and Implementation

Build both Li-Fi circuits on a breadboard, then solder them. Write firmware for OOK encoding and decoding, BLE, and Wi-Fi on the Pico. Build the Pi 4 pipeline, the Pico web server, and the KMP app with charts and notifications.

3.1.4 Testing Phase

Run unit tests for the codec and the benchmark calculation logic. Run integration tests for the MQTT to app data flow. Run benchmark trials across range, interference, and obstruction. Run power and temperature tests for each method in isolation.

Activities and estimated hours

The table below breaks the 280-hour budget down by activity, grouped by phase.

Phase	Activity	Hours
3.2.1 Requirements and Analysis	Literature review on Li-Fi, BLE, and Wi-Fi comparisons, OOK modulation, and benchmark methods	20
3.2.2 Design	Circuit and protocol design, included in the realization hours below, were tested on a breadboard before soldering.	
3.2.3 Realization and Implementation	Li-Fi link in both directions: LED, photodiode, amplifier, and comparator circuit on both sides, with firmware and logic for both directions	65
3.2.3 Realization and Implementation	Pico BLE and Wi-Fi firmware, including sending data and receiving the confirmation	20
3.2.3 Realization and Implementation	Pi 4 pipeline: aggregation, logging, benchmarking, confirmations on all three methods, MQTT publishing	40
3.2.3 Realization and Implementation	Power measurement setup with INA219 sensors, plus temperature comparison for each method in isolation	20
3.2.3 Realization and Implementation	Kotlin Multiplatform app: data models, MQTT client, state handling, UI structure	45
3.2.3 Realization and Implementation	Chart integration and notifications for each platform	15
3.2.3 Realization and Implementation	Pico web server for live verification of the current reading	10
3.2.4 Testing	Benchmark testing across range, interference, and power, with repeated trials	20
3.2.4 Testing and finalization	Report writing, analysis, reflection, and presentation	25
Total		280

4. Project Boundaries (scope and preconditions)

MoSCoW

Priority	Item
Must have	Li-Fi, BLE, and Wi-Fi transmit the same sensor reading at the same time, Pico 2 W to Pi 4. Confirmation in both directions on all three channels, including a Li-Fi return path that uses light Pi 4 benchmarking pipeline (timestamping, logging, throughput, BER, latency, power) publishing through MQTT Temperature and power test for each transmission method, run in isolation Report with schematics, source code, and analysis
Should have	Kotlin Multiplatform app with charts and notifications on Android, iOS, and Web or Desktop. Web server on the Pico for live verification of the current reading
Could have	Testing at more range and obstruction beyond the base test set iOS build of the KMP app, subject to device or Mac availability More interference scenarios, for example a congested 2.4 GHz Wi-Fi network
Will not have in this iteration	A Li-Fi network with more than one node or cell (this project uses one Pico to one Pi 4 link) A Li-Fi transceiver on the phone or app side Encryption of the sensor data An enclosure built for production or certification

Preconditions

- All listed components (Section 5.1) are available, including a spare Pico 2 W board.
- The supervisor, Mr. Christiaen Slot, is available for a meeting on request whenever a problem, question, or finding comes up.
- Access to a Saxion lab or workshop for soldering, circuit debugging, and testing.
- A room where lighting and radio conditions can be controlled is available for interference and obstruction tests.

5. Products

Hardware

Component	Qty	Purpose
Raspberry Pi 4	1	Receives all three transmission paths, uses an LED and photodiode for the Li-Fi return link, logs and benchmarks data, and publishes through MQTT
Raspberry Pi Pico 2 W, plus spare	1 plus 1	Sensor node: reads internal temperature, transmits over Li-Fi, BLE, and Wi-Fi, receives confirmations. Spare protects against damage during soldering or testing
LED	2	Li-Fi transmitters on the Pico and the Pi 4
Photodiode BPW34	2	Li-Fi receivers on the Pico and the Pi 4
Op amp, for example, LM358 or TL072	2	Transimpedance amplifier stage per photodiode circuit
Comparator LM393	2	Converts the amplified signal into a digital pulse
Resistor and capacitor kit	1	Gain and filtering for both amplifier circuits and other circuit needs
Breadboard and jumper wires	2 to 3 sets	Prototyping both Li-Fi circuits before soldering
INA219 current and voltage sensor	1	Measures Pico power use per transmission method
Small perfboard	2	Holds the soldered Li-Fi circuits
Enclosures	2	Protects the Pico and Pi 4 circuits
USB cables and power supplies	as needed	Powers the Pico boards and the Pi 4
Phone, Android and iOS if available	1 to 2	Runs the Kotlin Multiplatform app for live viewing and testing

Software

- Pico firmware: OOK Li-Fi encoder and decoder, BLE stack integration, Wi-Fi stack integration, web server.
- Raspberry Pi 4 aggregation and benchmark pipeline, including MQTT publishing.
- MQTT broker configuration.
- Kotlin Multiplatform application: shared data models, MQTT client, benchmark and state logic, and chart and notification integrations for Android, iOS, and Web or Desktop.

Documentation

- Report: source code, firmware, circuit schematics, and analysis.
- Project repository on GitHub with literature review, unit tests, and integration tests.
- A record of hours spent on the project.
- Presentation slides with a live demonstration, including a live interference test per method.

6. Quality assurance

Hardware

Before integration, the transimpedance amplifier gain and the comparator threshold are calibrated against a known light source. Solder joints on the perfboard are checked by eye and for continuity. INA219 readings are checked against a known load. Each benchmark condition runs repeated trials to control variance.

Software

Unit tests cover the codec and the benchmark calculation logic, tracked in the repository unit testing folder. Integration tests cover the MQTT to app data flow, tracked in the integration testing folder. Git tracks all code. The supervisor reviews implementation at each checkpoint.

Documentation

The report follows the structure of this project plan. The supervisor reviews it at each milestone. Every literature claim links to the References list (APA 7th edition). The repository README stays current. The hour log is updated weekly against the estimate in Section 3.2.

7. Project organization

Organization

Role	Name
Student	Van Dung Phan, 546821, 546821@student.saxion.nl
Supervisor	Mr. Christiaan Slot
Institution	Saxion University of Applied Sciences
Module	Capita Selecta, 10 EC

Information

- Progress is reported to the supervisor by email every week: what was done, what comes next, and progress against the plan in Chapter 8.
- A meeting with the supervisor happens only when needed, for a problem, a question, or a finding the student wants to discuss. There is no weekly meeting.
- The GitHub repository holds the code, documentation, and literature review.
- Hours are logged weekly in the repository against the estimate in Section 3.2.

8.Planning

The 280-hour budget spreads over the 16 active weeks of the 20-week module, about 17.5 hours per week. Weeks 17 to 20 are held as buffer and exam weeks, outside this budget:

Period	Focus
Weeks 1 to 3	Requirements and analysis: literature review, benchmark protocol and metric definitions (Section 3.2.1)
Weeks 3 to 9	Design and realization of the Li-Fi link in both directions, plus BLE and Wi-Fi firmware
Weeks 8 to 13	Pi 4 pipeline, Kotlin Multiplatform app, chart integration and notifications, Pico web server
Weeks 10 to 16	Power measurement setup, benchmark testing across range, interference, and obstruction, report writing, and presentation preparation
Weeks 17 to 20	Buffer, revision, and exam weeks, outside the 280-hour budget

Appendix A gives a breakdown of activity and week.

9. Risk analysis

Risk

9.1.1 Technical Risks

Risk	Mitigation
Ambient light interferes with the Li-Fi photodiode signal.	Shielding around the receiver, a comparator threshold that can be adjusted, and OOK framing with error checking
Wi-Fi and BLE radio coexistence on the Pico 2 W causes timing jitter, since both share one radio.	This is documented as a hardware characteristic. Firmware accounts for the coexistence of scheduling.
A board is damaged during soldering or testing.	A second Pico 2 W is kept as a spare.
KMP integration behaves differently across Android, iOS, and Web.	Core logic stays in shared code. Platform layers stay thin. Each platform gets a smoke test.
Internal temperature reading mixes CPU load and transmission load.	Each method is tested alone, with the other two idle, and checked against INA219 power readings.

9.1.2 Planning and Time Risks

Risk	Mitigation
The Li-Fi link is the largest work item, 65 of 280 hours, and carries schedule risk if circuit tuning takes longer than expected.	Start it first. Test on breadboard before soldering. Limit the time spent on each iteration.
Parallel EE and CS work could create an integration bottleneck near the end.	Define the MQTT message and topic contract early. Hardware and app work can then proceed on separate tracks.
iOS testing depends on device or Mac availability.	iOS is scoped as Should have or Could have (Section 4.1). Android and Web or Desktop come first.
Literature review or report writing scope creep.	Hour budget per activity (Section 3.2), and weekly written progress reports that show drift. A meeting is requested once a drift is noticed.

Conclusion

Project risk stays manageable. Each transmission method can be built and tested on its own, so a delay on one method, most likely Li-Fi, does not block the others. Mitigations include a spare Pico board, an hour budget split by phase, an early message contract between hardware and app, and weekly written progress reports that show issues before they grow. The MoSCoW boundaries in Chapter 4 back these mitigations: items such as the iOS build or extra interference scenarios can be dropped if schedule pressure appears. [OBJ]

10. References

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Appendix A: Detailed planning (Gantt overview)

Weeks are grouped into blocks of 4 weeks across the 20-week module. An X marks the blocks where an activity is the focus. Small amounts of follow-on work in nearby blocks are not shown. Weeks 17 to 20 are buffer and exam weeks, outside the 280-hour budget.

Activity	Week 1 to 4	Week 5 to 8	Week 9 to 12	Week 13 to 16	Week 17 to 20
Literature review & protocol def.	X				
Li-Fi circuit + firmware (65h)	X	X	X		
BLE / Wi-Fi firmware (20h)		X			
Pi 4 pipeline (40h)		X	X		
KMP app (45h)			X	X	
Charts and notifications (15h)			X	X	
Pico web server (10h)			X		
Power/INA219 setup (20h)			X		
Benchmark testing (20h)				X	
Report, analysis, presentation (25h)				X	
Buffer and exam weeks					X